

WASP-12b

R-0328

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231219 · created 2026-07-18T04:05:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460296.7733 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.011
Transit depth (Rp/Rs)^2	2.34 +/- 0.35 [%]
Orbital Inclination (inc)	81.59 +/- 2.61
Airmass coefficient 1 (a1)	1.052 +/- 0.016
Airmass coefficient 2 (a2)	-0.046 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	#2 - [205, 232]
Optimal Aperture	3.75
Optimal Annulus	12.96
Transit Duration (day)	0.1244 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.153 $\pm$ 0.011 vs 0.1178 (deviation 2.21 σ)
Duration fit vs published	0.1244 d vs 0.125 d (deviation 0.06 σ)
Mid-time O–C	-1572.8 min
Depth SNR	9.6
Residual scatter	1.54 %

Light curve

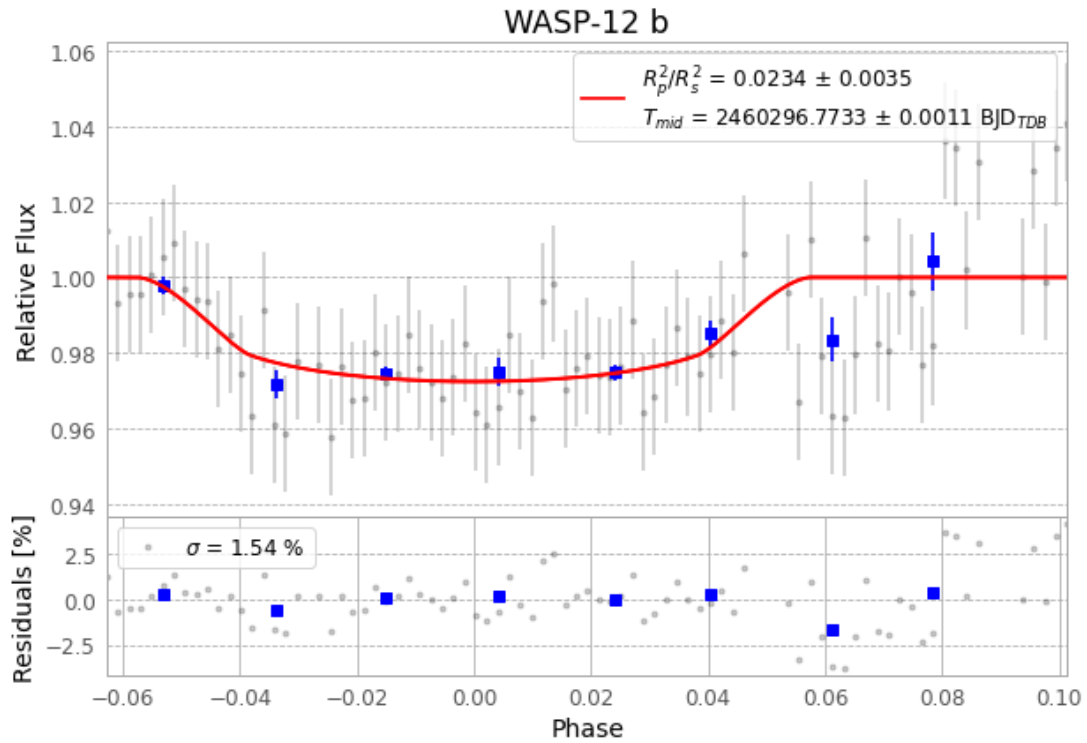


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-17.png (SHA-256 3431716c1f66cd27...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [317, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 56bc9f4623c56f3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

87 FITS frames (57 MB), WASP-12231218044515.FITS → WASP-12231218090315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231219.log (dc0cee71bf55...), FinalLightCurve_WASP-12 b_2023-12-17.png (3431716c1f66...), FinalLightCurve_WASP-12 b_2023-12-17.csv (ad9d4637ce75...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 04:05Z.